

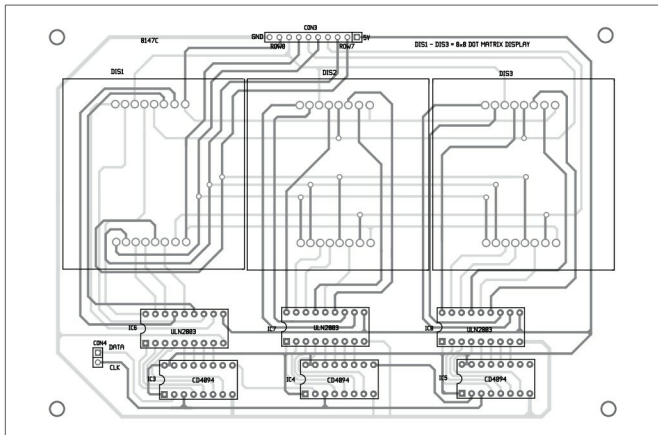


Fig. 9: Component-side track layouts of top as well as bottom layers of the display unit

the three shift registers, while data pulses are fed to pin 2 of the first shift register, IC3. The output of the first shift register from pin 9 is fed to pin 2 of the second shift register, IC4. To complete the cascading, output of the second shift register from its pin 9 is fed to pin 2 of the third shift register, IC5.

Data from the shift registers is used for activation or deactivation of columns of the three dot-matrix displays (DIS1-DIS3) through IC6-IC8 ICs. For controlling the logic levels at rows R0-R7, data from port P0 is transferred from the microcontroller to the display unit through connectors CON1 and CON3, with a supply of 5V and com-

mon ground to display unit of these two connectors.

Software

The source program for the display of 'EFY INDIA' is written in Assembly language and compiled using Keil μ Vision4 compiler. The generated hex code is burnt into the microcontroller using a suitable programmer.

Construction and testing

An actual-size, single-layer PCB layout for the power supply and the microcontroller unit is shown in Fig. 5 and its component layout in Fig. 6. PCB for the display unit is a double-sided

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

one. Track layout of its bottom layer is shown in Fig. 7 and of top layer in Fig. 8. Component and track layouts of the top layer as well as the bottom layer are shown in Fig. 9.

Assemble the circuits on the PCBs to save time and avoid assembly errors. ●

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